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ELECTRONIC DEVICE AND METHOD FOR OPERATING ELECTRONIC DEVICE

Abstract

A first chip includes a first substrate. The first substrate includes a first surface and a control circuit. A second chip includes a second substrate and an electrode. The second substrate includes a second surface facing the first surface. A plurality of bonding parts is located between the first surface and the second surface. One of the plurality of bonding parts electrically connects the control circuit and the electrode. The second substrate includes a diode electrically connected with the electrode and the plurality of bonding parts. A direction from the diode toward the electrode is a forward direction of the diode. A direction from the control circuit toward the diode via the one of the plurality of bonding parts is a reverse direction of the diode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-035008, filed on Mar. 7, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to an electronic device and a method for operating an electronic device.

BACKGROUND

[0003] A known device includes two chips bonded by a metal-to-metal bond.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIGS. 1 and 2 are schematic views showing a configuration of an electronic device of a first embodiment;

[0005] FIG. 3 is a schematic cross-sectional view showing a configuration of an electronic device of a second embodiment; and

[0006] FIG. 4 is a schematic plan view showing a configuration of a first surface side of a first substrate of the electronic device of the second embodiment.

DETAILED DESCRIPTION

[0007] According to one embodiment, an electronic device includes a first chip including a first substrate, the first substrate including a first surface and a control circuit; a second chip including a second substrate and an electrode, the second substrate including a second surface facing the first surface in a first direction; and a plurality of bonding parts located between the first surface and the second surface, one of the plurality of bonding parts electrically connecting the control circuit and the electrode, the plurality of bonding parts bonding the first chip and the second chip, the second substrate including a diode electrically connected with the electrode and the plurality of bonding parts, a direction from the diode toward the electrode being a forward direction of the diode, a direction from the control circuit toward the diode via the one of the plurality of bonding parts being a reverse direction of the diode.

[0008] Exemplary embodiments will now be described with reference to the drawings.

[0009] The drawings are schematic or conceptual; and the relationships between the thickness and width of portions, the proportional coefficients of sizes among portions, etc., are not necessarily the same as the actual values thereof. Furthermore, the dimensions and proportional coefficients may be illustrated differently among drawings, even for identical portions.

[0010] In the specification of the application and the drawings, components similar to those described in regard to a drawing thereinabove are marked with like reference numerals, and a detailed description is omitted as appropriate.

First Embodiment

[0011] As shown in FIG. 1, an electronic device 1 of a first embodiment includes a first chip 101, a second chip 102, and multiple bonding parts 40. The multiple bonding parts 40 are located between the first chip 101 and the second chip 102. The first chip 101 and the second chip 102 are bonded

to each other via the multiple bonding parts **40**, and are stacked in a first direction Z.

[0012] The first chip **101** includes a first substrate **10**. The first substrate **10** includes a first surface **10A**. The first substrate **10** also includes a control circuit **11** and a voltage write circuit **12**, which include semiconductor transistors formed at the first surface **10A** side. The control circuit **11** and the voltage write circuit **12** use the semiconductor transistors to perform switching functions. The voltage write circuit **12** may be located outside the first chip **101**. The first substrate **10** is, for example, a silicon substrate.

[0013] The second chip **102** includes a second substrate **20** and an electrode **70**. The second substrate **20** includes a second surface **20A** facing the first surface **10A** of the first substrate **10** in the first direction Z. The second substrate **20** is, for example, a silicon substrate.

[0014] The multiple bonding parts **40** bond the first chip **101** and the second chip **102** and are electrically connected with the control circuit **11**, the voltage write circuit **12**, and the electrode **70**.

[0015] The bonding parts **40** include first metal parts **41** and second metal parts **42**. The first metal parts **41** are located on the first surface **10A** and electrically connected with the control circuit **11** and the voltage write circuit **12**. The second metal parts **42** are located on the second surface **20A** and electrically connected with the electrodes **70**. The first metal part **41** and the second metal part **42** face each other in the first direction Z and are bonded to each other. The bond between the first metal part **41** and the second metal part **42** is, for example, a gold-gold bond.

[0016] A first external connection part **61** and a second external connection part **62** are located on the first surface **10A** of the first substrate **10**. The first external connection part **61** and the second external connection part **62** are located on the first surface **10A** in regions that do not face the second surface **20A** of the second substrate **20**. The first external connection part **61** is electrically connected with the control circuit **11**. The second external connection part **62** is electrically connected with the voltage write circuit **12**. The first external connection part **61** and the second external connection part **62** can be formed of the same material as the first metal part **41** by the same process as the first metal part **41**.

[0017] The second substrate **20** further includes a diode D. The diode D includes a p-n junction of a first semiconductor layer of a first conductivity type and a second semiconductor layer of a second conductivity type located in the second substrate **20**. In the specification, for example, the first conductivity type is taken to be a p-type; and the second conductivity type is taken to be an n-type. Or, the first conductivity type may be the n-type; and the second conductivity type may be the p-type.

[0018] The diode D is electrically connected with the electrode **70** and the bonding parts **40**. The anode of the diode D is electrically connected with a bonding part **40D** among the multiple bonding parts **40** that is electrically connected with the voltage write circuit **12**. The bonding part **40D** that is electrically connected with the voltage write circuit **12** is provided separately from the bonding part **40** electrically connected with the control circuit **11** and the electrode **70**. The cathode of the diode D is electrically connected with the electrode **70** and the second metal part **42**. The direction from the diode D toward the electrode **70** is the forward direction; and the direction from the control circuit **11** toward the diode D via the bonding part **40** is the reverse direction. The direction from the voltage write circuit **12** toward the diode D via the bonding part **40D** is the forward direction.

Normal Operation of Electronic Device

[0019] In a normal operation of the electronic device **1**, a positive operating voltage (e.g., 5 V) or 0 V is applied to the first external connection part **61**. The voltage that is applied to the first external connection part **61** is applied to the electrode **70** via the control circuit **11** and the bonding part **40**. In other words, the voltage of the electrode **70** of the second chip **102** can be controlled by the first chip **101**.

[0020] The electronic device **1** includes multiple first external connection parts **61**, multiple control circuits **11**, multiple bonding parts **40**, multiple electrodes **70**, and multiple diodes D, and includes multiple systems in which the individual first external connection part **61**, control circuit **11**,

bonding part **40**, electrode **70**, and diode **D** are electrically connected to each other in each system. At least one second external connection part **62**, at least one voltage write circuit **12**, and at least one bonding part **40D** may be included. Multiple bonding parts **40D** may be included. The second external connection part **62**, the voltage write circuit **12**, and the bonding part **40D** are electrically connected to the anodes of the multiple diodes **D**.

[0021] When a bonding part **40C** that has a bonding defect between the first metal part **41** and the second metal part **42** is present among the multiple bonding parts **40**, an electrode **70** is present that cannot be controlled to have the voltage applied to the first external connection part **61**. According to the embodiment, even when the bonding part **40C** that has the bonding defect is present, it is possible to adapt as described below.

Voltage Write Operation

[0022] A voltage write operation that is described below is performed before the normal operation of the electronic device **1**. First, as shown in FIG. **1**, the same positive writing voltage (e.g., 5 V) as the operating voltage described above is applied to the second external connection part **62**. The writing voltage is applied to the electrodes **70** via the voltage write circuit **12**, the bonding part **40D**, and the diodes **D**. The writing voltage is applied collectively to the multiple electrodes **70** via the multiple diodes **D**. The potentials of the multiple electrodes **70** are collectively written to be the desired potential (e.g., 5 V).

[0023] A monitor circuit that is located outside the electronic device **1** is connected to the first external connection part **61**. 0 V is applied to the first external connection part **61** via the monitor circuit in the state in which the writing voltage described above is applied to the electrode **70**; and the monitor circuit detects the current flowing through the control circuit **11**. The bonding goodness of the bonding part **40** can be determined based on the detection result.

[0024] The current from the second external connection part **62** flows through the control circuit **11** via the diode **D** and the bonding part **40** that has a good bond. When the bonding part **40C** that has a bonding defect is present, the current from the second external connection part **62** is blocked by the bonding part **40C** having the bonding defect and does not flow through the control circuit **11**. Based on the presence or magnitude of the current flowing through the control circuit **11**, the bonding part **40C** that has the bonding defect, i.e., the electrode **70** of which the voltage cannot be controlled by the control circuit **11**, can be identified.

[0025] After applying the writing voltage from the second external connection part **62** to the electrodes **70** via the voltage write circuit **12** and the diodes **D**, 0 V is applied to the second external connection part **62** while maintaining the state in which 0 V is applied to the first external connection parts **61** as shown in FIG. **2**. As a result, the potentials of the electrodes **70** connected to the bonding parts **40** having good bonds become 0 V via the bonding parts **40**, the control circuits **11**, the first external connection parts **61**, and the monitor circuit. The potential of the electrode **70** connected to the bonding part **40C** having the bonding defect becomes a floating potential; and the writing voltage described above is maintained.

[0026] In the normal operation after the write operation described above, the voltages of the electrodes **70** connected to the bonding parts **40** having good bonds can be controlled to be the operating voltage (e.g., 5 V) or 0 V according to the voltages applied to the first external connection parts **61**. In the normal operation, the voltage of the electrode **70** connected to the bonding part **40C** having the bonding defect is maintained at the voltage written in the write operation regardless of the voltage applied to the first external connection part **61**. The writing voltage is equal to the operating voltage in the normal operation, and so in the normal operation, the voltage of the electrode **70** connected to the bonding part **40C** having the bonding defect is maintained at the operating voltage (e.g., 5 V) regardless of the voltage applied to the first external connection part **61**. As a result, in the normal operation, the operation is not lost even for the electrode **70** connected to the bonding part **40C** having the bonding defect.

[0027] For example, the technology of the embodiment also is applicable to a sensor element of a

circuit reading data in which a switch is switched on when a voltage is applied, etc. In such an element, it is favorable for the switch to remain on so that the data can be accessed.

Second Embodiment

[0028] As shown in FIG. 3, an electronic device 2 of a second embodiment includes a first chip 201, a second chip 202, and the multiple bonding parts 40. The multiple bonding parts 40 are located between the first chip 201 and the second chip 202. The first chip 201 and the second chip 202 are bonded to each other via the multiple bonding parts 40, and are stacked in the first direction Z.

[0029] The first chip 201 includes the first substrate 10. The first substrate 10 includes the first surface 10A. The first substrate 10 also includes the control circuit 11 and the voltage write circuit 12 that include semiconductor transistors formed at the first surface 10A side. The control circuit 11 and the voltage write circuit 12 use the semiconductor transistors to perform switching functions. The voltage write circuit 12 may be located outside the first chip 201. The first substrate 10 is, for example, a silicon substrate.

[0030] The first chip 201 further includes a wiring part 30 located at the first surface 10A. The wiring part 30 includes an insulating layer 31, and a wiring layer located inside the insulating layer 31. The wiring layer includes a first wiring layer 32A and a second wiring layer 32B. For example, the wiring part 30 has a multilayer wiring structure in which wiring layers of different layers are electrically connected to each other by conductive vias.

[0031] The second chip 202 includes the second substrate 20, a first electrode 71, and a second electrode 72. The second substrate 20 includes a second surface 20A facing the wiring part 30 in the first direction Z. The second substrate 20 is, for example, a silicon substrate. The wiring part 30 is positioned between the first surface 10A and the second surface 20A in the first direction Z.

[0032] Multiple through-holes H extend through the second chip 202, the wiring part 30, and the first chip 201 in the first direction Z. For example, a ground potential is applied to the first and second substrates 10 and 20.

[0033] A first conductive film 91 and a second conductive film 92 are located at the second surface 20A of the second substrate 20 with an insulating film 50 interposed. The first electrode 71 and the second electrode 72 are located, with the insulating film 50 interposed, at the inner side surface of the portion of the through-hole H extending through the second substrate 20. The insulating film 50 is, for example, a silicon oxide film. The insulating film 50 is located between the second surface 20A and the first conductive film 91, between the second surface 20A and the second conductive film 92, between the first electrode 71 and the inner side surface of the through-hole H, and between the second electrode 72 and the inner side surface of the through-hole H.

[0034] The first electrode 71 and the second electrode 72 are separated from each other in the circumferential direction of the inner side surface at the inner side surface of the through-hole H. The first conductive film 91 and the second conductive film 92 are separated from each other on the insulating film 50. The first electrode 71 is electrically connected with the first conductive film 91. The second electrode 72 is electrically connected with the second conductive film 92. In the normal operation, mutually-different potentials are applied to the first and second electrodes 71 and 72.

[0035] A third conductive film 81, a fourth conductive film 82, a fifth conductive film 83, and a sixth conductive film 84 are arranged to be separated from each other on the wiring part 30. The third conductive film 81 and the fifth conductive film 83 are electrically connected with the control circuit 11 via the first wiring layer 32A and conductive vias of the wiring part 30. The fourth conductive film 82 and the sixth conductive film 84 are electrically connected with the voltage write circuit 12 via the second wiring layer 32B and conductive vias of the wiring part 30.

[0036] The multiple bonding parts 40 electrically connect the first chip 201 and the second chip 202 and bond the first chip 201 and the second chip 202. Each bonding part 40 includes metal parts that are located respectively at the first chip 201 side and the second chip 202 side and are bonded

to each other. The multiple bonding parts **40** include a first bonding part **40A** and a second bonding part **40B**. Multiple first bonding parts **40A** and multiple second bonding parts **40B** are located between the wiring part **30** and the second surface **20A** of the second chip **202**.

[0037] The first bonding part **40A** includes the first metal part **41** and the second metal part **42**. The first metal part **41** contacts the third conductive film **81** and is electrically connected with the control circuit **11** via the third conductive film **81**, conductive vias, and the first wiring layer **32A**. The second metal part **42** contacts the first conductive film **91** and is electrically connected with the first electrode **71** via the first conductive film **91**. The first metal part **41** and the second metal part **42** face each other in the first direction **Z** and are bonded to each other.

[0038] The second bonding part **40B** includes a third metal part **43** and a fourth metal part **44**. The third metal part **43** contacts the fourth conductive film **82** and is electrically connected with the voltage write circuit **12** via the fourth conductive film **82**, conductive vias, and the second wiring layer **32B**. The fourth metal part **44** contacts the second conductive film **92** and is electrically connected with the second electrode **72** via the second conductive film **92**. The third metal part **43** and the fourth metal part **44** face each other in the first direction **Z** and are bonded to each other.

[0039] The first external connection part **61** and the second external connection part **62** are located on the wiring part **30**. The first external connection part **61** and the second external connection part **62** are located on the wiring part **30** in regions that do not face the second surface **20A** of the second substrate **20**. The first external connection part **61** contacts the fifth conductive film **83** and is electrically connected with the control circuit **11** via the fifth conductive film **83**, conductive vias, and the first wiring layer **32A**. The second external connection part **62** contacts the sixth conductive film **84** and is electrically connected with the voltage write circuit **12** via the sixth conductive film **84**, conductive vias, and the second wiring layer **32B**. The first external connection part **61** and the second external connection part **62** can be formed of the same material as the first and third metal parts **41** and **43** in the same process as the first and third metal parts **41** and **43**.

[0040] The second substrate **20** further includes the diode **D**. Multiple diodes **D** are arranged in the second substrate **20** to correspond to the multiple first electrodes **71**. The diode **D** includes a p-type first semiconductor layer **21** (a p-type well) located in the second substrate **20**, and an n-type second semiconductor layer **22** located inside the first semiconductor layer **21**. The first semiconductor layer **21** and the second semiconductor layer **22** have a p-n junction. The first semiconductor layer **21** functions as the anode of the diode **D**; and the second semiconductor layer **22** functions as the cathode of the diode **D**.

[0041] The second semiconductor layer **22** is electrically connected with the first conductive film **91** by a conductive via extending through the insulating film **50** located on the second surface **20A**. Accordingly, the cathode of the diode **D** is electrically connected with the first bonding part **40A** via the first conductive film **91**.

[0042] The second substrate **20** can include a p-type third semiconductor layer **23** that is located inside the first semiconductor layer **21** and has a higher p-type impurity concentration than the first semiconductor layer **21**. The third semiconductor layer **23** functions as a contact layer of the diode **D** at the anode side. The third semiconductor layer **23** is electrically connected with the second conductive film **92** by a conductive via extending through the insulating film **50** located on the second surface **20A**. Accordingly, the anode of the diode **D** is electrically connected with the second bonding part **40B** via the second conductive film **92**.

[0043] The anode of the diode **D** (the first semiconductor layer **21**) is electrically connected with the second external connection part **62** via the second bonding part **40B**, the second wiring layer **32B**, and the voltage write circuit **12**. The cathode of the diode **D** (the second semiconductor layer **22**) is electrically connected with the first electrode **71** and the first bonding part **40A**. The direction from the diode **D** toward the first electrode **71** via the first conductive film **91** is the forward direction. The direction from the control circuit **11** toward the diode **D** via the first bonding part **40A** is the reverse direction. The direction from the voltage write circuit **12** toward the diode **D**

via the second bonding part **40B** is the forward direction.

[0044] The electronic device **2** of the second embodiment can be used as a pixel array that deflects electron beams of a multi-electron beam lithography apparatus that uses multiple electron beams to draw a pattern in a resist on a substrate of glass, etc. The electron beam passes through the through-hole **H**. The electron beam that passes through the through-hole **H** can be deflected by the electric field generated between the first electrode **71** and the second electrode **72**.

[0045] By bonding two chips (the first chip **201** and the second chip **202**) that each have through-holes, the electronic device **2** that has a deep through-hole that would be difficult to form in a solitary chip can be configured.

[0046] Normal operation of electronic device In the normal operation of the electronic device **2**, a positive operating voltage (e.g., 5 V) or 0 V is applied to the first external connection part **61**. The voltage that is applied to the first external connection part **61** is applied to the first electrode **71** via the first wiring layer **32A**, the control circuit **11**, the first bonding part **40A**, and the first conductive film **91**.

[0047] 0 V is applied to the second external connection part **62**. 0 V that is applied to the second external connection part **62** is applied to the second electrode **72** via the second wiring layer **32B**, the voltage write circuit **12**, the second bonding part **40B**, and the second conductive film **92**.

[0048] When 0 V is applied to the first and second electrodes **71** and **72**, the electron beam that passes through the through-hole **H** is not deflected, and is irradiated on the drawing object. When 0 V is applied to the second electrode **72** and the operating voltage (e.g., 5 V) is applied to the first electrode **71**, the electron beam that passes through the through-hole **H** is deflected, and is not irradiated on the drawing object.

[0049] When the bonding part **40C** that has a bonding defect between the first metal part **41** and the second metal part **42** is present among the multiple first bonding parts **40A**, the operating voltage cannot be applied to the first electrode **71** that is to be connected with the bonding part **40C** having the bonding defect; and the electron beam that passes through the through-hole **H** cannot be deflected. This results in a defect in which the electron beam is undesirably irradiated at a position at which drawing is not to be performed. According to the embodiment, even when the bonding part **40C** having the bonding defect is present, it is possible to adapt as described below.

Voltage Write Operation

[0050] The voltage write operation described below is performed before the electronic device **2** performs the normal operation (the drawing operation of the electron beam on the drawing object). First, the same positive writing voltage (e.g., 5 V) as the operating voltage described above is applied to the second external connection part **62**. The writing voltage is applied to the first electrodes **71** via the voltage write circuit **12**, the second wiring layer **32B**, the second bonding part **40B**, the second conductive film **92**, the diodes **D**, and the first conductive films **91**. The writing voltage is collectively applied to the multiple first electrodes **71** via the multiple diodes **D**. The potentials of the multiple first electrodes **71** are collectively written to the desired potential (e.g., 5 V).

[0051] A monitor circuit that is located outside the electronic device **2** is connected to the first external connection part **61**. 0 V is applied to the first external connection part **61** via the monitor circuit in the state in which the writing voltage described above is applied to the first electrode **71**; and the current that flows through the control circuit **11** is detected by the monitor circuit. The bonding goodness of the first bonding part **40A** can be determined based on the detection result.

[0052] The current from the second external connection part **62** flows through the control circuits **11** via the diodes **D**, the first conductive films **91**, and the first bonding parts **40A** having good bonds. When the bonding part **40C** that has a bonding defect is present, the current from the second external connection part **62** is blocked by the bonding part **40C** having the bonding defect, and does not flow through the control circuit **11**. The bonding part **40C** that has the bonding defect, i.e., the first electrode **71** of which the voltage cannot be controlled by the control circuit **11**, can be

identified based on the presence or magnitude of the current flowing through the control circuit **11**.
[0053] After applying the writing voltage from the second external connection part **62** to the first electrodes **71** via the voltage write circuit **12**, the second bonding part **40B**, and the diodes **D**, 0 V is applied to the second external connection part **62** while maintaining a state in which 0 V is applied to the first external connection parts **61**. As a result, the potentials of the first electrodes **71** connected to the first bonding parts **40A** having good bonds are set to 0 V via the first conductive films **91**, the first bonding parts **40A**, the first wiring layer **32A**, the control circuits **11**, the first external connection parts **61**, and the monitor circuit. The potential of the first electrode **71** connected to the bonding part **40C** having the bonding defect becomes a floating potential; and the writing voltage described above is maintained.

[0054] In the normal operation after the write operation described above, the voltage of the first electrode **71** connected to the first bonding part **40A** having the good bond can be controlled to be the operating voltage (e.g., 5 V) or 0 V according to the voltage applied to the first external connection part **61**. In the normal operation, the voltage of the first electrode **71** connected to the bonding part **40C** having the bonding defect is maintained at the voltage written in the write operation, regardless of the voltage applied to the first external connection part **61**. The writing voltage is equal to the operating voltage in the normal operation, and so in the normal operation, the voltage of the first electrode **71** connected to the bonding part **40C** having the bonding defect is maintained at the operating voltage (e.g., 5 V) regardless of the voltage applied to the first external connection part **61**. As a result, even when the bonding part **40C** that has a bonding defect is present, the electron beam that passes through the through-hole **H** at which the first electrode **71** connected to the bonding part **40C** having the bonding defect is located can be deflected, and a defect of the electron beam being undesirably irradiated on a position at which drawing is not to be performed can be avoided.

[0055] In the normal operation, the first electrode **71** that is connected to the bonding part **40C** having the bonding defect can no longer be controlled to be 0 V. Therefore, the electron beam that passes through the through-hole **H** (the defective pixel) at which the first electrode **71** connected to the bonding part **40C** having the bonding defect is located can no longer be irradiated on the drawing object.

[0056] Although it is desirable to avoid defects in which the electron beam is undesirably irradiated at a position at which drawing is not to be performed, there are cases where a prescribed number of defects in which the electron beam is not irradiated at the position to be drawn are allowed. The latter defect can be remedied using other normal pixels positioned around the defective pixel that includes the first electrode **71** connected to the bonding part **40C** having the bonding defect.

[0057] For example, in a technique in which one pixel is formed by irradiating an electron beam multiple times, the electron beam is irradiated, the mask is then moved slightly, then the irradiation of the electron beam is repeated, and the defective pixel can be remedied by using multiple electron beams to form one pixel.

[0058] If the voltage of the first electrode **71** connected to the bonding part **40C** having the bonding defect drops due to current leakage, etc., the voltage of the first electrode **71** connected to the bonding part **40C** having the bonding defect can be maintained at the operating voltage by performing the write operation described above again. For example, the drop of the voltage of the first electrode **71** connected to the bonding part **40C** having the bonding defect can be detected by monitoring the deflection amount of the electron beam.

[0059] FIG. **4** is a schematic plan view showing an example of a configuration of the electronic device **2** at the first surface **10A** side of the first substrate **10**. In FIG. **4**, a second direction **X** and a third direction **Y** are orthogonal to each other in a plane parallel to the first surface **10A**. The second direction **X** and the third direction **Y** also are orthogonal to the first direction **Z**.

[0060] The multiple through-holes **H** are arranged in the second and third directions **X** and **Y**. The third conductive films **81** are arranged with a one-to-one correspondence to the through-holes **H**.

The first bonding parts **40A** are arranged in contact with the third conductive films **81** with a one-to-one correspondence with the through-holes H. One through-hole H is positioned between the first bonding parts **40A** adjacent to each other in the third direction Y. One through-hole H is positioned between the second bonding parts **40B** adjacent to each other in the second direction X. Two second bonding parts **40B** are provided for one through-hole H and arranged with the through-hole H interposed in the second direction X. The number of the second bonding parts **40B** is greater than the number of the first bonding parts **40A**. The potentials of the multiple second electrodes **72** can be collectively controlled via the multiple second bonding parts **40B** and the common fourth conductive film **82**. The potentials of the multiple first electrodes **71** are individually controlled via the corresponding first bonding part **40A** and the corresponding third conductive film **81**.

[0061] The multiple second bonding parts **40B** are located on the common fourth conductive film **82** and contact the fourth conductive film **82**. As shown in FIG. 3, the multiple second bonding parts **40B** have a common connection with the first semiconductor layers **21** that are the anodes of the diodes D. Accordingly, even when the second bonding part **40B** that has a bonding defect is present among the multiple second bonding parts **40B**, the writing voltage can be collectively applied to the multiple first electrodes **71** via the diodes D in the write operation described above via the other second bonding parts **40B** having good bonds.

[0062] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modification as would fall within the scope and spirit of the inventions.

Claims

1. An electronic device, comprising: a first chip including a first substrate, the first substrate including a first surface and a control circuit; a second chip including a second substrate and an electrode, the second substrate including a second surface facing the first surface in a first direction; and a plurality of bonding parts located between the first surface and the second surface, one of the plurality of bonding parts electrically connecting the control circuit and the electrode, the plurality of bonding parts bonding the first chip and the second chip, the second substrate including a diode electrically connected with the electrode and the plurality of bonding parts, a direction from the diode toward the electrode being a forward direction of the diode, a direction from the control circuit toward the diode via the one of the plurality of bonding parts being a reverse direction of the diode.
2. The electronic device according to claim 1, wherein the diode includes: a first semiconductor layer located in the second substrate, the first semiconductor layer being of a first conductivity type; and a second semiconductor layer located inside the first semiconductor layer, the second semiconductor layer being of a second conductivity type.
3. The electronic device according to claim 2, wherein the second chip includes a conductive film located on the second surface, and the conductive film is electrically connected with the second semiconductor layer, the electrode, and the one of the plurality of bonding parts.
4. The electronic device according to claim 1, further comprising: a through-hole extending through the second and first substrates in the first direction, the electrode being a first electrode, the second chip including a second electrode, the first electrode and the second electrode being located at an inner side surface of a portion of the through-hole extending through the second substrate, the plurality of bonding parts including a first bonding part electrically connected with the control circuit, the first electrode, and a cathode of the diode, and a second bonding part electrically

connected with the second electrode and an anode of the diode.

5. The electronic device according to claim 4, wherein the first electrode and the second electrode are separated from each other in a circumferential direction of the inner side surface at the inner side surface of the through-hole.
 6. The electronic device according to claim 1, wherein the first chip includes a wiring part located between the first surface and the second surface, and the wiring part includes a wiring layer electrically connected with the control circuit and the one of the plurality of bonding parts.
 7. The electronic device according to claim 4, wherein the first chip includes a wiring part located between the first surface and the second surface, and the wiring part includes a wiring layer electrically connected with the control circuit and the first bonding part.
 8. The electronic device according to claim 7, wherein the wiring layer is a first wiring layer, the wiring part further includes a second wiring layer, the first wiring layer is electrically connected with the first electrode, and the second wiring layer is electrically connected with the second electrode.
 9. The electronic device according to claim 1, wherein the first chip further includes an external connection part located on the first surface of the first substrate in a region not facing the second surface of the second substrate, and the external connection part is electrically connected with the control circuit.
 10. The electronic device according to claim 1, wherein the first substrate is a silicon substrate, and the control circuit includes a semiconductor transistor.
 11. A method for operating the electronic device according to claim 1, the method comprising: applying a voltage from a voltage write circuit to the electrode via the diode; and applying 0 V to the plurality of bonding parts after the applying of the voltage to the electrode.
 12. The method according to claim 11, further comprising: determining a bonding goodness of at least one of the plurality of bonding parts based on a current flowing through the control circuit in a state in which the voltage is applied to the electrode.
 13. The method according to claim 11, wherein the electronic device further includes a through-hole extending through the second and first substrates in the first direction, the electrode is a first electrode, the second chip includes a second electrode, the first electrode and the second electrode are located at an inner side surface of a portion of the through-hole extending through the second substrate, the plurality of bonding parts includes: a first bonding part electrically connected with the control circuit, the first electrode, and a cathode of the diode; and a second bonding part electrically connected with the second electrode and an anode of the diode, and the method further comprises: applying the voltage to the first electrode via the second bonding part and the diode.
 14. The method according to claim 13, wherein mutually-different potentials are applied to the first and second electrodes.
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